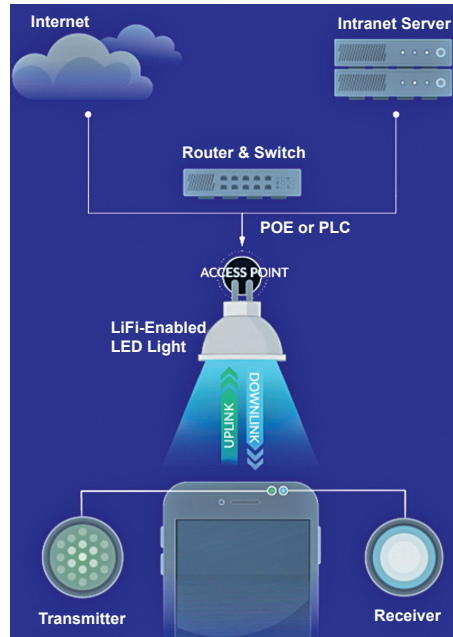


Li-Fi—the world's fastest Wi-Fi

Li-Fi is a wireless internet communication system that uses visible light for transmission at very high speeds. Combining the world of electronics and photonics, this technology captures data in modulated light frequencies of an LED light source and transmits them. The photosensitive detector converts the light frequency signal back to an electronic data stream, allowing faster bi-directional wireless communication.

Like decoding Morse code, this technology has a much faster rate of decoding information, which is up to millions of times per second. In terms of transmission speed, Li-Fi can go over 100Gbps, which is nearly 14 times faster than even WiGig.

The basics of a Li-Fi system
(Credit: <https://Li-Fi.co/what-is-Li-Fi>)



Using Li-Fi technology, Sonam Wangchuk conducted a live Youtube session on April 1, 2022 and displayed the speed of the internet.

In terms of geography, the signal from the Jio tower cannot be received nor seen at the SECMOL school. However, it can be seen and received on the SECMOL mountain (Li-Fi Tower1 hub), which is at an altitude of 800m. After the signal is captured at the Li-Fi Tower1 hub as a line-of-sight (LOS) propagation from the Jio tower, it transmits the signal to the optical communication unit (Figure 6) via light modulation (laser beams). This unit is placed on the roof of the SECMOL school and the signal from the Li-Fi Tower1 hub is decoded and converted back to an electrical signal. Hence the signal from the tower can reach remote areas (indirectly), as seen in Figure 6.

Li-Fi eliminates the cost for digging of optical fibre cables and building of 10-story tall towers. However, powering the towers can be a challenge considering the cold atmosphere on the mountain top. Major telecom providers say that the power required for each tower is up to 7kW. Optical units on the other hand require only 48W.

Just like the solar-heated building, Navtech, HIAL, and SECMOL have developed a solar-heated battery storage device. When the temperatures in the barrel (below the solar panel) were measured, a difference of approx. 61 degrees was obtained. Sonam says, “The temperatures in the night are a minimum of +26 degrees, even while it is a chilling -40 degrees outside. This proved the power system can work unmanned on mountain-tops.”

As shown in Figure 4, using Li-Fi receivers and transmitters, any signal (example, telecom) can be obtained from a normal tower. When a laser-emitting device is installed on the top of the mountain, any remote village below

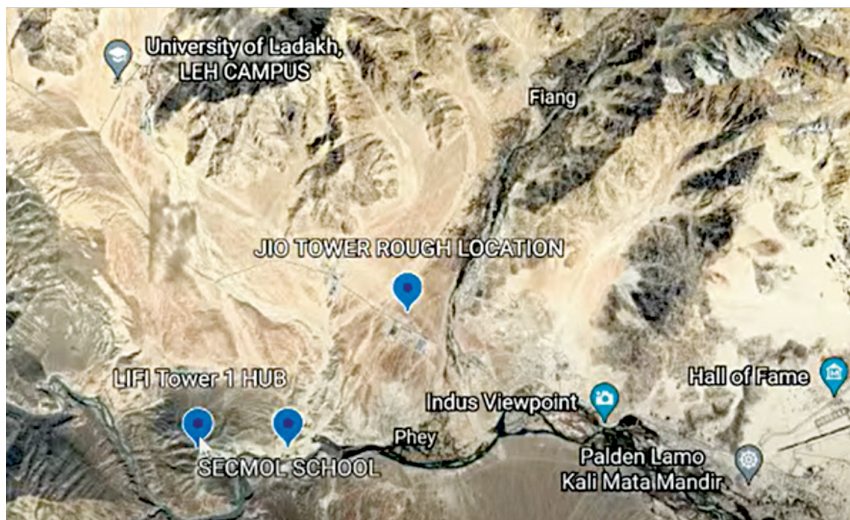


Figure 4: Geographical location



Figure 5: Optical communication unit for Li-Fi

Mountain-to-mountain connectivity

The top of a mountain is usually isolated and cut-off with no internet connectivity found. But mountains are the mightiest and biggest structures that can be used to erect towers. Combining this with an interesting technology called Li-Fi, the mountain-to-mountain connectivity can be achieved.